

Report: An Analysis of Wasserstein GAN with Gradient Penalty (WGAN-GP)

Author: Mohammadmahdi Bababeyk

Student ID: 4041419005

1.0 Theoretical Motivation for Wasserstein GANs

The training of standard Generative Adversarial Networks (GANs) is notoriously difficult, often suffering from critical issues such as training instability and the vanishing gradients problem. These challenges significantly hinder the generator's ability to learn and produce high-quality synthetic data. In response, researchers developed more robust frameworks, with the Wasserstein GAN (WGAN) emerging as a pivotal advancement that directly addresses the foundational weaknesses of earlier architectures.

Limitations of Standard GANs

An analysis of a standard Deep Convolutional GAN (DCGAN) trained on the FashionMNIST dataset reveals the typical pathologies associated with this architecture. The training logs and loss curves highlight several key issues:

- **Discriminator Dominance:** The discriminator's loss is observed to frequently approach zero. For instance, the training log shows the loss dropping to values as low as $D_Loss=0.0192$ by Epoch 5 Batch 200. This indicates that the discriminator has become overwhelmingly proficient at its task, easily distinguishing between real and fake images and breaking the delicate equilibrium required for effective adversarial training.
- **Vanishing Gradients:** A direct and debilitating consequence of a near-perfect discriminator is the vanishing gradients problem. When the discriminator's loss is approximately zero, it provides no meaningful gradient signal for the generator to learn from. The loss function saturates, creating a plateau where the generator receives no useful feedback to guide its improvement.
- **Training Instability:** The generator's loss curve for the DCGAN does not decrease steadily but instead exhibits high volatility and occasional large spikes. The initial iterations are particularly illustrative, with the generator's loss jumping from approximately 2.18 to over 12.8 within the first 100 batches. This erratic behavior is a clear indicator of training instability, where

the generator struggles to adapt to a constantly shifting and overly powerful discriminator.

The Earth Mover's Distance Solution

The Wasserstein GAN introduces a fundamental change to the training objective to circumvent these problems. Its core innovation is the use of the Wasserstein distance, also known as the Earth Mover's (EM) distance, as a more robust metric for comparing the real and generated data distributions.

To achieve this, WGAN replaces the standard discriminator—a binary classifier outputting a probability—with a "critic." The critic's role is not to classify but to estimate the Wasserstein distance between the two distributions. The key benefit of this approach is that the Wasserstein distance provides a "more meaningful and stable loss signal," which is less prone to saturation. This stability stems from the fact that the Wasserstein distance is a continuous and differentiable metric everywhere, even when the two distributions do not overlap. This provides a smoother loss landscape, which is the technical reason it prevents the gradient from vanishing.

This theoretical shift necessitates several key modifications to the model's implementation, transforming both the network architecture and the loss function.

2.0 Key Architectural and Implementation Differences

The theoretical advantages of WGANs are realized through a series of specific and critical modifications to the network architecture, loss function, and training procedure. An examination of the provided WGAN-GP implementation reveals the following key changes compared to a standard DCGAN.

- **From Discriminator to Critic**
 - The primary conceptual change is the replacement of the Discriminator with a WassersteinCritic. A standard discriminator is a binary classifier that outputs a probability (typically between 0 and 1) indicating whether an image is real or fake. In contrast, the critic is designed to perform a regression-like task, outputting a continuous, un-bounded score used to estimate the Wasserstein distance.
- **Removal of Sigmoid Activation**

- To enable the output of an un-bounded score, the final `nn.Sigmoid()` activation function must be removed from the critic's architecture. The source implementation of the `WassersteinCritic` class correctly omits this final layer, which is present in the standard `Discriminator` class. This allows the network to output any real number, which is essential for approximating the Wasserstein distance.
- **Wasserstein Loss Implementation**
 - The `nn.BCELoss()` (Binary Cross-Entropy Loss) used in the DCGAN is replaced with a loss function that reflects the Wasserstein objective. As implemented in the training loop, the core loss for the critic is calculated as: $-(\text{torch.mean}(\text{critic_real}) - \text{torch.mean}(\text{critic_fake}))$. This formulation encourages the critic to output higher scores for real images and lower scores for fake images, thereby maximizing the distance between their mean scores, which serves as the approximation of the Wasserstein distance.
- **Enforcing the 1-Lipschitz Constraint via Gradient Penalty**
 - A critical requirement for the critic's output to be a valid approximation of the Wasserstein distance is that the critic function must be 1-Lipschitz continuous. The source material implements the recommended Gradient Penalty (GP) method to enforce this constraint. This is achieved through the `get_gradient_penalty` function, which calculates a penalty term based on the norm of the critic's gradients with respect to its inputs. This penalty, weighted by the hyperparameter `LAMBDA_GP`, is then added directly to the critic's loss, encouraging the model to adhere to the constraint.

These architectural modifications directly enable the superior training dynamics that characterize the WGAN-GP framework.

3.0 Training Stability Analysis

A primary objective in generative modeling research is the development of reliable metrics that can diagnose and guide the training process. The WGAN-GP framework provides such a metric, leading to a more stable and interpretable training dynamic

compared to its predecessors. The core advantages of this approach stem directly from its use of the Wasserstein distance.

The "Bonus" section of the source material highlights several key benefits that contribute to this enhanced stability:

- **Meaningful Loss Correlation:** The critic's loss serves as a meaningful, though not always monotonically decreasing, proxy for perceptual sample quality. As the generator improves, the Wasserstein distance estimate between the real and generated distributions should shrink, a trend that can be directly monitored, unlike the often uninterpretable loss of a standard GAN.
- **Reduced Mode Collapse:** The WGAN objective function inherently encourages the generator to produce a greater diversity of samples. This helps mitigate mode collapse, a common failure case in GANs where the generator learns to produce only a few distinct types of outputs that can successfully fool the discriminator, ignoring the full diversity of the training data.
- **Prevention of Vanishing Gradients:** As previously discussed, the Wasserstein loss does not saturate in the same way as the cross-entropy loss used with a sigmoid-based discriminator. This property is fundamental to WGAN's stability, as it ensures that the generator always receives a usable, informative gradient, even when the critic is highly effective.

The empirical results from the experimental training run provide clear evidence supporting these theoretical claims.

4.0 Experimental Observations

To empirically validate the theoretical claims of improved stability, a WGAN-GP model was trained for 10 epochs on the FashionMNIST dataset. The training logs provide clear, observable evidence that the WGAN-GP framework behaves as expected.

The critic's loss (Loss C) exhibits a distinct and informative pattern throughout the training process. This dynamic behavior is a hallmark of a healthy WGAN-GP training process; it indicates that the critic is not saturating but is actively adapting and providing a continuous, responsive gradient signal to the generator. This stands in stark contrast to the "stalled" or "plateaued" state a DCGAN discriminator enters

when its loss hits zero. The training log confirms this, showing the critic loss reaching its most negative value of -16.5623 early in training (Epoch 0, Batch 100) and settling to values like -6.9444 by the final epoch, demonstrating a persistent and active gradient signal throughout.

The observation that the critic's loss remains active and provides a continuous, non-saturating signal strongly supports the claim that WGAN-GP effectively mitigates the vanishing gradients problem. This provides the generator with a consistent learning signal, which is a cornerstone of the architecture's stability. These empirical findings align perfectly with the theoretical merits of the WGAN-GP framework.

5.0 Conclusion

This report has analyzed the theoretical motivations, key implementation changes, and empirically observed performance of the Wasserstein GAN with Gradient Penalty (WGAN-GP) as a significant improvement over standard GAN architectures. The analysis of both the DCGAN and WGAN-GP training processes reveals that the WGAN-GP framework successfully addresses the most critical training pathologies that have long challenged the field of generative modeling.

The core findings of this analysis confirm that WGAN-GP is a superior training paradigm for the following reasons:

1. **Resolves Critical Training Pathologies:** WGAN-GP directly addresses the vanishing gradient problem that plagues standard GANs. By replacing the sigmoid-based classifier with a critic and a non-saturating gradient, it ensures a reliable learning process.
2. **Provides an Interpretable Proxy for Sample Quality:** The critic's loss serves as a meaningful proxy for image quality, a feature notably absent in standard GANs where loss values are often uninterpretable. This allows for more effective monitoring and debugging of the training process.
3. **Enhances Generation Quality and Stability:** By promoting stable gradients and reducing the likelihood of mode collapse, the WGAN-GP framework facilitates the generation of more diverse and higher-quality samples, enabling more principled and predictable training of deep generative models.

Ultimately, the WGAN-GP architecture represents a significant and practical advancement in the field of generative modeling, offering a more stable and principled approach to training deep generative models.